Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

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Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



07/07/2026

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Session	2025-26 (ODD)	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	22	Name of Student	Sanskruti . P . Kharapkar

Practical Number	7
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a Weather App using HTML, CSS, and JavaScript that fetches live weather information for a user-entered city using the Open-Meteo API.
Problem Definition	To create a web application that allows users to enter a city name and view its current temperature, humidity, wind speed, and weather code. The application should fetch data from the Open-Meteo Geocoding and Forecast APIs, handle errors, and display results dynamically without reloading the webpage.
Theory (100 words)	Asynchronous programming allows JavaScript to perform operations such as API requests without blocking the webpage. The `fetch()` function sends HTTP requests and returns a Promise. The `async` and `await` keywords make asynchronous code easier to write and understand. In this project, the Geocoding API converts a city name into latitude and longitude, which are used by the Weather API to retrieve current weather data.

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Procedure and Execution

(100 Words)

Step for Implementation:

1. **Create a folder named `WeatherApp`.**
2. **Create `index.html`, `style.css`, and `script.js`.**
3. **Design the page using HTML and CSS.**
4. **Use an event listener to handle the Search button.**
5. **Fetch city coordinates and weather details using Open-Meteo APIs.**
6. **Display the results and handle errors using `try...catch`.**
7. **Run the application in a browser and test different city names.**

Code:

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>ShopEase | Online Shopping</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js" defer></script>
</head>
<body>
  <header class="site-header">
    <a class="brand" href="#top" aria-label="ShopEase
home"><span class="brand-mark">S</span> ShopEase</a>
    <p class="tagline">Find your everyday favourites</p>
    <a class="cart-link" href="#cart-section"
aria-label="Jump to your cart">
      <span aria-hidden="true">🛒</span> Your Cart <span
class="cart-count" id="cart-count">0</span>
    </a>
  </header>

  <main id="top" class="page-wrap">
    <section class="intro">
```

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```
      <p class="eyebrow">SHOP THE COLLECTION</p>
      <h1>Explore Items</h1>
      <p class="intro-copy">Useful picks, everyday
essentials, and little upgrades for you.</p>
    </section>

    <section aria-label="Available products">
      <div id="product-list" class="product-grid"
aria-live="polite"></div>
    </section>

    <section id="cart-section" class="cart-section">
      <div class="section-heading">
        <div>
          <p class="eyebrow">YOUR SELECTION</p>
          <h2>Your Shopping Cart <span
class="muted-count" id="cart-item-count">(0
items)</span></h2>
        </div>
        <button id="clear-cart" class="text-button"
type="button" hidden>Clear cart</button>
      </div>
      <div id="cart-list" class="cart-list"
aria-live="polite"></div>
    </section>

    <section id="bill-section" class="bill-section">
      <div class="bill-card">
        <div class="bill-title-row">
          <div>
            <p class="eyebrow">ORDER TOTAL</p>
            <h2>Bill Summary</h2>
          </div>
          <span class="bill-icon"
aria-hidden="true">▤</span>
        </div>
```

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```
        <div
class="bill-line"><span>Subtotal</span><strong
id="subtotal">₹0</strong></div>
        <div class="bill-line"><span>Delivery
charge</span><strong id="delivery">₹0</strong></div>
        <div class="bill-line
discount-line"><span>Discount</span><strong
id="discount">-₹0</strong></div>
        <div class="bill-line total-line"><span>Total
amount</span><strong id="total-price">₹0</strong></div>
        <p class="delivery-note" id="delivery-note">Add
items to see your order total.</p>
        <button id="checkout" class="checkout-button"
type="button">Proceed to Checkout <span
aria-hidden="true">→</span></button>
        <p id="message" class="checkout-message"
role="status" aria-live="polite"></p>
        <p class="secure-note"><span
aria-hidden="true">🔒</span> Demo checkout · No payment
will be collected</p>
      </div>
    </section>
  </main>

  <footer>ShopEase · Student JavaScript Practical
Project</footer>
</body>
</html>
```

JAVASCRIPT

```
const cityInput = document.getElementById("cityInput");
const searchButton =
document.getElementById("searchButton");
const weatherResult =
document.getElementById("weatherResult");
```

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```
searchButton.addEventListener("click", () => {
    const city = cityInput.value.trim();

    if (city === "") {
        showMessage("Please enter a city.");
        cityInput.focus();
        return;
    }

    getWeather(city);
});

cityInput.addEventListener("keydown", (event) => {
    if (event.key === "Enter") {
        searchButton.click();
    }
});

async function getWeather(city) {
    searchButton.disabled = true;
    searchButton.textContent = "Searching...";
    weatherResult.innerHTML = `
        <div class="message-card">
            <div class="empty-icon"
aria-hidden="true">⏳</div>
            <h2>Loading weather...</h2>
            <p>Getting the latest available weather for
${escapeHTML(city)}.</p>
        </div>`;

    try {
        // Request 1: Convert the city name into latitude
and longitude.
        const geoUrl =

"https://geocoding-api.open-meteo.com/v1/search?name=" +
```

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```
            encodeURIComponent(city) +
"&count=1&language=en&format=json";

        const geoResponse = await fetch(geoUrl);
        if (!geoResponse.ok) {
            throw new Error("Geocoding request failed");
        }

        const geoData = await geoResponse.json();

        if (!geoData.results || geoData.results.length
=== 0) {
            showMessage("City not found. Please check the
spelling and try again.");
            return;
        }

        const location = geoData.results[0];

        // Request 2: Use the coordinates to get the
current weather.
        const weatherUrl =
            "https://api.open-meteo.com/v1/forecast" +
            "?latitude=" + location.latitude +
            "&longitude=" + location.longitude +

"&current=temperature_2m,relative_humidity_2m,apparent_te
mperature,weather_code,wind_speed_10m" +
            "&timezone=auto";

        const weatherResponse = await fetch(weatherUrl);
        if (!weatherResponse.ok) {
            throw new Error("Weather request failed");
        }

        const weatherData = await weatherResponse.json();
```

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```
        if (!weatherData.current) {
            throw new Error("Current weather data is
unavailable");
        }

        displayWeather(location, weatherData.current);
    } catch (error) {
        console.error("Weather App error:", error);
        showMessage("Unable to fetch weather data. Check
your internet connection and try again.");
    } finally {
        searchButton.disabled = false;
        searchButton.textContent = "Search";
    }
}

function displayWeather(location, current) {
    const condition =
getWeatherCondition(current.weather_code);
    const placeParts = [location.admin1,
location.country].filter(Boolean);
    const placeDetails = placeParts.join(", ");
    const updatedAt = current.time
        ? "Local time: " + current.time.replace("T", " ")
        : "Current conditions";

    weatherResult.innerHTML = `
        <article class="weather-card">
            <div class="weather-top">
                <div>
                    <p class="location-label">Current
weather</p>
                    <h2>${escapeHTML(location.name)}</h2>
                    <p
class="conditions">${escapeHTML(placeDetails)}</p>
                    <p
class="conditions">${condition.description}</p>
```

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```
                </div>
                <div class="weather-symbol"
aria-hidden="true">${condition.icon}</div>
            </div>
            <div class="temperature-row">
                <div
class="temperature">${current.temperature_2m}°C</div>
                <div class="feels-like">
                    <strong>Feels like
${current.apparent_temperature}°C</strong><br>
                    Temperature at the location
                </div>
            </div>
            <div class="metrics">
                <div class="metric">
                    <div class="metric-icon"
aria-hidden="true">💧</div>
                    <p class="metric-label">Humidity</p>
                    <p
class="metric-value">${current.relative_humidity_2m}%</p>
                </div>
                <div class="metric">
                    <div class="metric-icon"
aria-hidden="true">💨</div>
                    <p class="metric-label">Wind
speed</p>
                    <p
class="metric-value">${current.wind_speed_10m} km/h</p>
                </div>
                <div class="metric">
                    <div class="metric-icon"
aria-hidden="true">🌡️</div>
                    <p class="metric-label">Weather
code</p>
                    <p
class="metric-value">${current.weather_code}</p>
                </div>
```

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```
            </div>
            <p
class="updated-time">${escapeHTML(updatedAt)} · Data by
Open-Meteo</p>
        </article>`;
}

function getWeatherCondition(code) {
    if (code === 0) return { description: "Clear sky",
icon: "☀️" };
    if ([1, 2].includes(code)) return { description:
"Mainly clear / partly cloudy", icon: "🌤️" };
    if (code === 3) return { description: "Overcast",
icon: "☁️" };
    if ([45, 48].includes(code)) return { description:
"Foggy", icon: "🌫️" };
    if ([51, 53, 55, 56, 57].includes(code)) return {
description: "Drizzle", icon: "🌦️" };
    if ([61, 63, 65, 66, 67].includes(code)) return {
description: "Rain", icon: "🌧️" };
    if ([71, 73, 75, 77].includes(code)) return {
description: "Snow", icon: "❄️" };
    if ([80, 81, 82].includes(code)) return {
description: "Rain showers", icon: "🌧️" };
    if ([85, 86].includes(code)) return { description:
"Snow showers", icon: "🌨️" };
    if ([95, 96, 99].includes(code)) return {
description: "Thunderstorm", icon: "⛈️" };
    return { description: "Current conditions", icon:
"🌡️" };
}

function showMessage(message) {
    weatherResult.innerHTML = `
        <div class="message-card error-card">
            <h2>Weather update</h2>
            <p>${escapeHTML(message)}</p>
```

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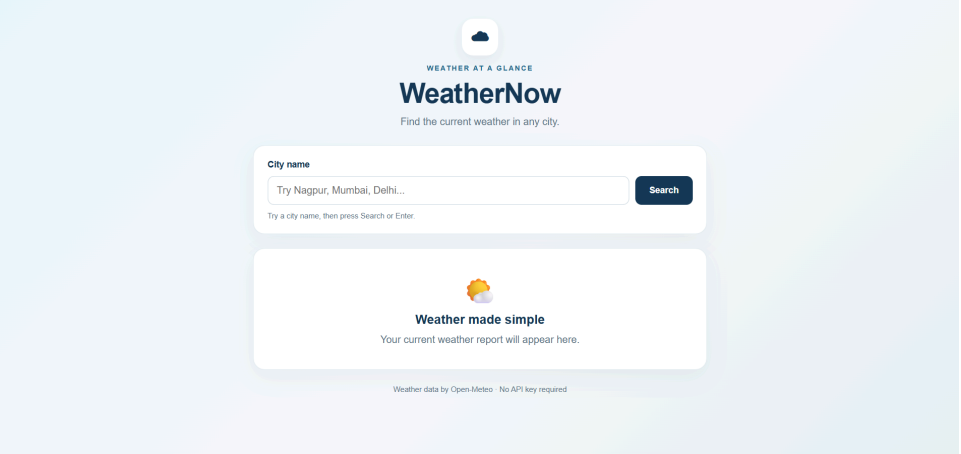
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```
        </div>`;
}

// Keep user-entered text safe when it is displayed in
the page.
function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, (character)
=> ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        '"': "&quot;",
        "'": "&#39;"
    }[character]));
}
```

Output:

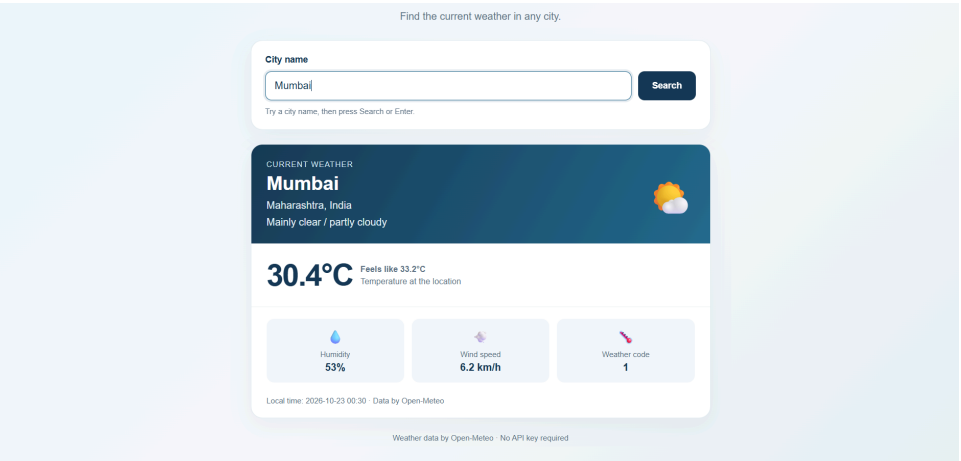


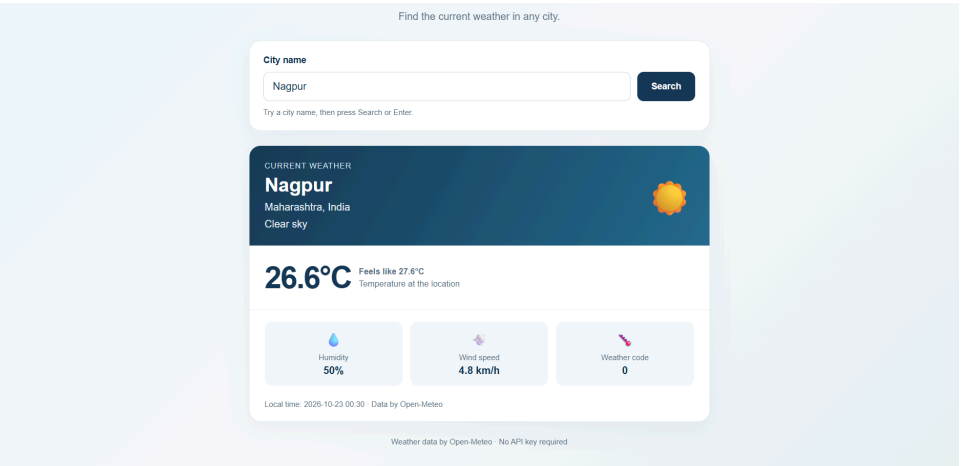
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<table>
<tr><td></td><td> 


</td></tr>
<tr><td>Output Analysis</td><td>The application successfully demonstrates asynchronous JavaScript programming using the Open-Meteo API. The <code>fetch()</code> function retrieves data, while <code>async/await</code> waits for API responses without blocking the webpage. The Geocoding API provides coordinates, which are used to fetch weather details. The DOM is updated dynamically with the received data. Input validation and error handling improve the application's reliability.</td></tr>
<tr><td>Link of student Github profile where lab assignment has been uploaded</td><td>Sanskruti2007-code/web-Technology-Practicals</td></tr>
</table>

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Conclusion	**The Weather App demonstrates how HTML, CSS, and JavaScript can be used to build an interactive web application. It provides live weather information through API requests and applies asynchronous programming concepts such as `fetch()`, Promises, `async/await`, JSON parsing, and error handling. This practical improves understanding of API integration and dynamic DOM updates.**
Plag Report (Similarity index < 12%)	**9%**
Date	**28/09/2026**